

Organising the mathematical model into input parameters, operations/ equations and output parameters

1 . Input parameters

1.1 Physical Drone parameters

Mass

Arm Length

I_{xx} I_{yy} I_{zz}

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

1.2 Initial States of the drone $[x,y,z,dx,dy,dz,\Phi,\Theta,\Psi,p,q,r]^T$

$[x,y,z]$ = The initial coordinates of the drone in the 3D space.

$[dx,dy,dz]$ = The initial linear velocities

$[\Phi,\Theta,\Psi]$ = The initial Euler angles

$[p,q,r]$ = The initial angular velocities

1.3 Desired states of the drone

$$\mathbf{r} = \begin{bmatrix} \phi_{des} \\ \theta_{des} \\ \psi_{des} \\ \dot{z}_{des} \end{bmatrix}.$$

The desired Euler angles and the vertical velocities which are passed on from the controller.

1.4 PID gain constants

Kp_phi, Ki_phi, Kd_phi

Kp_theta, Ki_theta, Kd_theta

Kp_psi, Ki_psi, Kd_psi

Kp_zdot, Ki_zdot, Kd_zdot

2. Output parameters

Lets list out the output parameters before going to the operations

2.1 Velocities and accelerations of the drone (instantaneous)

$$\dot{\mathbf{x}} = [\dot{X}, \dot{Y}, \dot{Z}, \ddot{X}, \ddot{Y}, \ddot{Z}, \dot{\phi}, \dot{\theta}, \dot{\psi}, \dot{p}, \dot{q}, \dot{r}]^T$$

All the differentials of the first input parameters are obtained. Therefore the X dot matrix has the velocities and accelerations

2.2 Controller generated inputs

$$\mathbf{u} = \begin{bmatrix} T \\ M_1 \\ M_2 \\ M_3 \end{bmatrix}$$

The Thrust and the moments about the three axes are calculated using PID formulas which later help in determining the accelerations listed above.

3. Operations and Equations

3.1 Controller generated inputs

Using the PID control constants and errors, the controller calculates the controller generated inputs. (listed in the matrix $\mathbf{u}$)

The following terms are being used in the equations:

For roll:

$$e_\phi = \phi_{des} - \phi$$

For pitch:

$$e_\theta = \theta_{des} - \theta$$

For yaw:

$$e_\psi = \psi_{des} - \psi$$

For vertical velocity:

$$e_z = \dot{z}_{des} - \dot{z}$$

The conventional PID formula is modified towards a discrete version.

The general PID equation used is:

$$u = K_P e + K_I \int e \, dt + K_D \frac{de}{dt}$$

Your discrete implementation is:

$$u_k = K_P e_k + K_I \sum_{i=0}^k e_i + K_D \frac{e_k - e_{k-1}}{dt}$$

Roll controller

$$M_1 = K_{P\phi} e_\phi + K_{I\phi} \sum e_\phi + K_{D\phi} \frac{e_\phi - e_{\phi,prev}}{dt}$$

This produces the roll moment M_1 .

Pitch controller

$$M_2 = K_{P\theta} e_\theta + K_{I\theta} \sum e_\theta + K_{D\theta} \frac{e_\theta - e_{\theta,prev}}{dt}$$

This produces M_2 .

Yaw controller

$$M_3 = K_{P\psi} e_\psi + K_{I\psi} \sum e_\psi + K_{D\psi} \frac{e_\psi - e_{\psi,prev}}{dt}$$

This produces M_3 .



$$T = K_{Pz} e_{\dot{z}} + K_{Iz} \sum e_{\dot{z}} + K_{Dz} \frac{e_{\dot{z}} - e_{\dot{z},prev}}{dt}$$

3.2 Formulation of a rotational matrix

A rotational matrix R is continuously calculated at every instant dt to define the orientation of the body frame with respect to the inertial frame of reference.

Rotation about X-axis

$$R_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}$$

Rotation about Y-axis

$$R_2(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

Rotation about Z-axis

$$R_3(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Your code then calculates:

$$R = R_1 R_2 R_3$$

3.3 Drone's linear acceleration

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \frac{1}{m} \left[\begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix} + R \begin{bmatrix} 0 \\ 0 \\ -T \end{bmatrix} \right]$$

Where R and T are defined earlier.

3.4 Drone's angular acceleration

The angular velocity vector is:

$$\boldsymbol{\omega} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}.$$

The moments generated by the controller are:

$$\mathbf{M} = \begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix}.$$

Your code uses the Newton-Euler equation:

$$\mathbf{I}\dot{\boldsymbol{\omega}} = \mathbf{M} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})$$

Therefore:

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1} [\mathbf{M} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega})]$$

The resulting column matrix lists down the three components of the angular acceleration

$$\dot{p}, \dot{q}, \dot{r}$$

3.5 Rate of change of Euler angles

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Where p,q,r are the components of the angular velocity of the drone.

Finally this gives us the matrix:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \ddot{x} \\ \ddot{y} \\ \ddot{z} \\ \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix}.$$

These generated outputs direct account for the drone simulation